BFKL evolution of the projectile: all terms $\propto \log \frac{\vee}{k^+}$ from your `BFKL.tex` and `Other_terms.tex`

Notation as in your files; $\mathcal{N} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}$, $K(a,b,z) = \mathcal{K}(a-z)\cdot\mathcal{K}(b-z)$, $U^{bc} = U^{cb}$. In the rows: black = your `BFKL.tex`, blue = added (the $\log \frac{\vee}{k^+}$ still inside `Other_terms.tex`), red = to be removed. The remaining terms are in `Remaining_rows.pdf`.

1 Answer

- **Your two files are complete.** Row by row, `BFKL.tex` + `Other_terms.tex` = the row minus its $\log \frac{k^+}{\Lambda}$ (JIMWLK) part, numerically (the differences, $\le 2.5\times10^{-9}$ at $\Lambda = 10^{-9}$, are the $O(\Lambda)$ remainders of the $\log \frac{k^+}{\Lambda}$ lines); the terms with an unevaluated p^+, ξ or momentum integral are identical to those of your Group files. The only exceptions are the two red lines below.
- **Many BFKL terms are still in `Other_terms.tex`**: every logarithm there that grows with $\vee$ contains $\log \frac{\vee}{k^+}$,

$$\log\Big(\frac{\vee}{k^+}-1\Big) = \log\frac{\vee}{k^+} + \log\Big(1-\frac{k^+}{\vee}\Big), \qquad \log\frac{(\vee-k^+)A+k^+B}{\Lambda A+k^+B} = \log\frac{\vee}{k^+} + \log\frac{(1-\frac{k^+}{\vee})A+\frac{k^+}{\vee}B}{B} + O(\Lambda), \qquad \dots$$

  ($A=(x-z)^2$, $B=(x-w)^2$; the full table is Sec. 2 of `LL_terms.pdf`). They are added in blue: Group I Rows III and IV, Group II Row I (both parts), Group III Rows I and II. The exceptions are the hidden logarithms (Group III Rows III, V, VI and Group II Row IV), where $\log(\frac{\vee}{k^+}-1)$ cancels against $\left[\frac{1}{p^+}\right]_+$ (resp. $\frac{\xi}{(1-\xi)_+}$); there the BFKL term is (B) (resp. H_4), added in blue for Group III.
- **Remove two lines** (red):
  - Group I Row V, 2b UV: Row V has no $\log \frac{\vee}{k^+}$; this line comes from the missing factor 2 after "Setting $\Lambda\to 0$" (E1).
  - Group II Row IV, line 1, $-\log \frac{\vee}{k^+}\times$(pole bracket): this is the $\log \frac{\vee}{k^+}$ hidden in $\frac{\xi}{(1-\xi)_+}$; it cancels against the one in $\log(\frac{\vee}{k^+}-1)$ of line 2 of `Other_terms.tex`, so it is not a BFKL term.

  Their poles cancel each other (to 1.5×10^{-15}), but their sum is not zero (0.23 relative): it would add $\log \frac{\vee}{k^+}\log\frac{(w-w')^4}{(x-w)^2(x'-w')^2}$ times the LO structure, which is not in BFKL⊗LO (Sec. 9).
- **With these changes:**
  - every row of the BFKL list equals the $\log \frac{\vee}{k^+}$ list of `LL_terms.pdf` (to 4.4×10^{-17});
  - the 35 three-ρ terms $\propto \log \frac{\vee}{k^+}$ (rows + your d^3N_a, d^3N_b) **cancel**, to 3.5×10^{-16} (Sec. 6). Without the blue terms (your `BFKL.tex` alone, 9 terms) they leave 0.39;
  - your four-ρ two-ρ lists d^3N_1, d^3N_2 are equal to the ones in the checked sum (3.2×10^{-15});
  - the two-ρ terms add up to **BFKL⊗LO** exactly (9 kernel structures, 109 colour structures, 0 differences; Sec. 7);
  - the result is your (commented) "BFKL contribution", both forms, to 2.1×10^{-16}; in colour-singlet form it is your App. E formula (Sec. 8).

So

$$\frac{d^3N_{\text{BFKL}}}{d^3k} = \mathcal{N}\log\frac{\vee}{k^+}\,\text{BFKL}\otimes\text{LO}, \tag{1.1}$$

Eq. (8.5). The remaining terms (`Remaining_rows.pdf`) are your `Other_terms.tex` minus the blue terms here, with $\Lambda\to 0$, the subtractions $-$(B) and $-\log \frac{\vee}{k^+}H_4$, and C4, C5.

2 Group I (with c.c.)

2.1 Row I

Only four ρ's (Sec. 5).

2.2 Row II

Row II, first part. No BFKL term: $\log(1-\frac{k^+}{\vee})\log \frac{\vee}{k^+}\to 0$ for $\vee\to\infty$.

Row II, second part, d^3N_{2a}. As you have it. Your `BFKL.tex`:

$$\begin{aligned}\frac{d^3N_{2\,a}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\frac{(w-z)\cdot(y-z)}{(w-z)^2\,(y-z)^2}\right.\\ &\left.-\,\frac{(y-w)^i}{(y-w)^2}\,\frac{(w-z)\cdot(x-z)}{(w-z)^2\,(x-z)^2}\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\Big]\\ &-\,\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^4}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\log\frac{(w-y)^2}{(x-y)^2}-\frac{(y-w)^i}{(y-w)^2}\,\log\frac{(w-x)^2}{(x-y)^2}\right]\times\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\Big]\end{aligned}\tag{2.1}$$

Row II, second part, d^3N_{2b}. No BFKL term (R^i depends on $\vee$ only through $\frac{k^+}{\vee}$).

2.3 Row III

Row III, d^3N_1. Your line is correct. Added: lines 1, 2 and 4 of d^3N_1 in `Other_terms.tex` still contain $\log\frac{\vee}{k^+}$: $\log(\frac{\vee}{k^+}-1)=\log\frac{\vee}{k^+}+\log(1-\frac{k^+}{\vee})$, $\log\frac{\vee(x-z)^2-k^+(x-w)^2}{(\Lambda+k^+)(x-z)^2-k^+(x-w)^2}=\log\frac{\vee}{k^+}+\ldots$, $\log\frac{k^+(x-w)^2-\vee(x-z)^2}{k^+(x-w)^2-\Lambda(x-z)^2}=\log\frac{\vee}{k^+}+\ldots$. Your `BFKL.tex`:

$$\begin{aligned}\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = &\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\end{aligned}\tag{2.2}$$

From `Other_terms.tex`:

$$\begin{aligned}\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = &\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{\\ &-\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-z)^2+(x-w)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ &+\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-w)^2+(x-z)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}\times\log\Big(\frac{\vee}{k^+}\Big)\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ &-\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\Big(\frac{\vee}{k^+}\Big)\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\}\end{aligned}\tag{2.3}$$

Row III, d^3N_2. Your line is correct. Added: the $\log\frac{\vee}{k^+}$ parts of all eight lines of d^3N_2 in `Other_terms.tex`. Your `BFKL.tex`:

$$\begin{aligned}\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = &\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,(x-w)^2}\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\end{aligned}\tag{2.4}$$

From `Other_terms.tex`:

$$\begin{aligned}\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{\\ &\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\big[(x-w)^2-(x-z)^2\big]^2}\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ &-\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\big[(x-w)^2-(x-z)^2\big]^2}\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ &+\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\big[(x-w)^2-(x-z)^2\big]}\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ &-\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\Big(\frac{\vee}{k^+}\Big)\,\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\big[(x-w)^2-(x-z)^2\big]\,(x-w)^2}\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\end{aligned}$$

$$
\begin{aligned}
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-z)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\}
\end{aligned}
\tag{2.5}
$$

Row III, d^3N_{3a}. No BFKL term.

Row III, d^3N_{3b}. No BFKL term.

2.4 Row IV

Row IV. Your line is correct. Added: the $\log\frac{\vee}{k^+}$ parts of $\log\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{\Lambda(x-z)^2+k^+(x-w)^2}$ and $\log(\frac{\vee}{k^+}-1)$ (both lines in `Other_terms.tex`). Your `BFKL.tex`:

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k}=&-\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,w,x',w'}e^{-ik\cdot(w'-w)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\times\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x)-f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\Big]
\end{aligned}
\tag{2.6}
$$

From `Other_terms.tex`:

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k}=&\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik\cdot(w'-w)}\Bigg\{\\
&\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left(\frac{\vee}{k^+}\right)\times\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.\\
&\left.-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}+\frac{\delta^{ij}}{2\,(z-w)^2}\right]\\
&+\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x)-f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\Big]
\end{aligned}
\tag{2.7}
$$

2.5 Row V

Row V, first part. No BFKL term.

Row V, second part, d^3N_{2a}. No BFKL term.

Row V, second part, $d^3N_{2b\,\rm UV}$. Remove (red). Row V has no $\log\frac{\vee}{k^+}$. Before "Setting $\Lambda\to 0$" your bracket is $2\log\frac{\vee}{\Lambda}-2\log\frac{\vee}{k^+}+2\log(1-\frac{\Lambda}{k^+})-\frac{11}{6}=2\log\frac{k^+}{\Lambda}+2\log(1-\frac{\Lambda}{k^+})-\frac{11}{6}$. The $2\log\frac{k^+}{\Lambda}$ is in your JIMWLK terms, and $2\log(1-\frac{\Lambda}{k^+})-\frac{11}{6}$ is in `Other_terms.tex`. Nothing is left for BFKL; this line comes from the bracket $\left[\log\frac{\vee}{k^+}+\log\frac{k^+}{\Lambda}\right]-2\log\frac{\vee}{k^+}$ after "Setting $\Lambda\to 0$", which lacks a factor 2 on the first bracket (E1). Its pole cancels the pole of the red line of Group II Row IV, Sec. 9. Your `BFKL.tex`:

$$
\begin{aligned}
\frac{d^3N_{2b\,\rm UV\,NLO}}{d^3k}=&-\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{y,z,x',w'}e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\\
&\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y).
\end{aligned}
\tag{2.8}
$$

Row V, second part, $d^3N_{2b\,\rm finite}$. No BFKL term.

3 Group II

3.1 Row I

Row I, first part, first line. Not in your `BFKL.tex`. Added: the log $\frac{\vee}{k^+}$ parts of all three lines (`Other_terms.tex`). From `Other_terms.tex`:

$$\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{ & \\
& -\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\log\left(\frac{\vee}{k^+}\right)\times \\
& \left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right] \\
& +\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2} \\
& -\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\} \\
& \left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned} \tag{3.1}$$

Row I, first part, second line. Added: the log $\frac{\vee}{k^+}$ parts of all three lines. From `Other_terms.tex`:

$$\begin{aligned}
\frac{d^3N}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{ & \\
& \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\log\left(\frac{\vee}{k^+}\right)\times \\
& \left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right] \\
& -\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w)\cdot(y-z)}{(x'-w)^2\,(y-z)^2\,(z-w')^2} \\
& +\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\} \\
& \times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right]
\end{aligned} \tag{3.2}$$

Row I, second part, d^3N_{2a}. No BFKL term: $\vee$ cancels between numerator and denominator of the first logarithm.

Row I, second part, d^3N_{2b}. Not in your `BFKL.tex`; all three lines are added. From `Other_terms.tex`:

$$\begin{aligned}
\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = & \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times \\
& \left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right. \\
& \left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times \\
& \left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\right. \\
& \left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\right] \\
& -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times \\
& \left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right. \\
& \left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left(\frac{\vee}{k^+}\right)\times \\
& \left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\right.
\end{aligned}$$

$$
- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Big]
$$

$$
+ \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-z)^2}{[(x'-w')^4\,(x-z)^2 \,-(x-w)^2\,(x'-z)^2\,(x-w')^2]} \times
$$

$$
\Big[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} + \frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}
$$

$$
+ \frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2} + \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2} \Big] \times \log\Big(\frac{\vee}{k^+}\Big)
$$

$$
\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x)
$$

$$
- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Big] \tag{3.3}
$$

Row I, second part, d^3N_{2c}. Not in your `BFKL.tex`; both lines are added. From `Other_terms.tex`:

$$
\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-w')^2}{(x'-z)^2}\, \frac{1}{[(x'-w')^2\,(x-z)^2 - (x-w)^2\,(x'-z)^2]}
$$

$$
\Big[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2} - \frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}
$$

$$
+ \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2} \Big] \times \log\Big(\frac{\vee}{k^+}\Big)
$$

$$
\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x)
$$

$$
- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Big]
$$

$$
- \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-w)^2}{(x-z)^2}\, \frac{1}{[(x'-w')^2\,(x-z)^2 - (x-w)^2\,(x'-z)^2]}
$$

$$
\Big[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2} - \frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}
$$

$$
+ \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2} \Big] \times \log\Big(\frac{\vee}{k^+}\Big)
$$

$$
\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x)
$$

$$
- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Big] \tag{3.4}
$$

Row I, second part, d^3N_{2d}. Your line is correct. Added: the two lines of `Other_terms.tex`. Your `BFKL.tex`:

$$
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+}\, \log\Big[\frac{\vee}{k^+}\Big] \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{1}{[(x-z)^2-(x-w)^2]}\, \frac{1}{[(x'-w')^2-(x'-z)^2]}
$$

$$
\Big[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}
$$

$$
+ \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2} \Big]
$$

$$
\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x)
$$

$$
- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \Big] \tag{3.5}
$$

From `Other_terms.tex`:

$$
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-z)^2}{[(x-z)^2-(x-w)^2]}\, \frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\Big[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}
$$

$$
+ \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2} \Big] \times
$$

$$
\begin{aligned}
&\log\left(\frac{\vee}{k^+}\right)\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\Big[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\\
&+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Big]\times\\
&\log\left(\frac{\vee}{k^+}\right)\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{3.6}
$$

Row I, second part, d^3N_{2e}. Your line is correct. Added: the two lines of `Other_terms.tex`. Your `BFKL.tex`:

$$
\begin{aligned}
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k}&=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}\\
&\Big[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\\
&+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Big]\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{3.7}
$$

From `Other_terms.tex`:

$$
\begin{aligned}
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k}&=-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\Big[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\\
&+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Big]\times\\
&\log\left(\frac{\vee}{k^+}\right)\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-w')^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\Big[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\\
&+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Big]\times\\
&\log\left(\frac{\vee}{k^+}\right)\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{3.8}
$$

Row I, second part, d^3N_{2f}. Your line is correct. Added: lines 1 and 3 of `Other_terms.tex`. Your `BFKL.tex`:

$$\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\log\left[\frac{\vee}{k^+}\right]\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{\left[(x'-w')^2\,(x-w)^2-(x-z)^2\,(x'-z)^2\right]}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]\,\left[(x'-z)^2-(x'-w')^2\right]}$$
$$\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x')-f^{ac'd'}\,U^{bc'}(z)U^{e'd'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right] \tag{3.9}$$

From `Other_terms.tex`:

$$\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]}\,\frac{\left[(x'-z)^2-(x'-w')^2\right]}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times$$
$$\log\left(\frac{\vee}{k^+}\right)\times$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]$$
$$+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}$$
$$\frac{\left[(x-z)^2-(x-w)^2\right]}{\left[(x'-z)^2-(x'-w')^2\right]\,\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times\log\left(\frac{\vee}{k^+}\right)\times$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big] \tag{3.10}$$

3.2 Row II

Only four ρ's (Sec. 5).

3.3 Row III

Only four ρ's (Sec. 5).

3.4 Row IV

Row IV. Line 2 (H_4) is correct. It is the BFKL term of this row, contained in the ξ integral of line 1 of `Other_terms.tex`; there it must be subtracted (Remaining terms, C1). **Line 1: remove (red).** It is the $-\log\frac{\vee}{k^+}$ hidden in the subtraction of $\frac{\xi}{(1-\xi)_+}$ in line 1 of `Other_terms.tex`. That one is cancelled exactly by the $+\log\frac{\vee}{k^+}$ inside $\log(\frac{\vee}{k^+}-1)$ of line 2 of `Other_terms.tex`. Line 1 of `Other_terms.tex` still contains it, so writing it here as well counts it twice. Together with the red line of Group I Row V it is pole free, but the two leave a term that is not in BFKL⊗LO (Sec. 9). Your `BFKL.tex`:

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = -\frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}$$
$$\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)$$
$$+\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}$$
$$\left[U^{e'c}(z)-U^{e'c}(x')\right]\times\left[U^{ec}(z)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x) \tag{3.11}$$

4 Group III (with c.c.)

4.1 Row I

Row I. Your line is correct. Added: the $\log\frac{\vee}{k^+}$ parts of both lines of `Other_terms.tex`. Your `BFKL.tex`:

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}$$
$$\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y)-U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right] \tag{4.1}$$

From `Other_terms.tex`:

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{ - \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left(\frac{\vee}{k^+}\right)\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.$$
$$\left. - \frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2} - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]$$
$$- \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times$$
$$\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{4.2}$$

4.2 Row II

Row II. Your line is correct. Added: the log $\frac{\vee}{k^+}$ parts of both lines of `Other_terms.tex`. Your `BFKL.tex`:

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{ \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]\Bigg\} \tag{4.3}$$

From `Other_terms.tex`:

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{ - \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left(\frac{\vee}{k^+}\right)\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.$$
$$\left. - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}\right]$$
$$- \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times$$
$$\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{4.4}$$

4.3 Row III

Row III, first part. Not in your `BFKL.tex`. Added: the term (B), the log $\frac{\vee}{k^+}$ hidden in the $-\frac{1}{k^+}$ term of line 1 (region $y'\to z$; `Finite_terms_rows.pdf`, Sec. 2). The $\log(\frac{\vee}{k^+}-1)$ of line 2 is *not* a BFKL term: it cancels the log $\frac{\vee}{k^+}$ of the subtraction in $\left[\frac{1}{p^+}\right]_+$. From `Other_terms.tex`:

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]$$
$$\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \tag{4.5}$$

Row III, second part. Added: (B) of this part (from the $-\frac{p^+}{k^{+2}}$ term of line 1). From `Other_terms.tex`:

$$\frac{d^3N_{2\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,\frac{(x'-z)\cdot(x-z)}{(x'-z)^2\,(x-z)^2}$$
$$\times\left[\frac{(y'-z)\cdot(z-w)}{(y'-z)^2\,(z-w)^2} - \frac{(y'-z)\cdot(x-w)}{2\,(y'-z)^2\,(x-w)^2}\right]$$
$$\left[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]$$
$$\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{4.6}$$

4.4 Row IV

Only four ρ's (Sec. 5).

4.5 Row V

Row V. Added: (B), as in Row III. From `Other_terms.tex`:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(z)\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\,\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\Big] \tag{4.7}$$

4.6 Row VI

Row VI. Added: (B), as in Row III. From `Other_terms.tex`:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]$$
$$\times\,\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\Big] \tag{4.8}$$

5 Four ρ's

Your `BFKL.tex`, unchanged: the three-ρ remainders d^3N_a, d^3N_b (they enter Sec. 6) and the two-ρ lists d^3N_1, d^3N_2. Numerically, $d^3N_1+d^3N_2$ equals the four-ρ two-ρ terms that entered the checked BFKL sum (Eqs. (7.13), (7.14) below) to 3.2×10^{-15} (charges symmetrized, with c.c.).

$$\frac{d^3N_{a\,\rm NLO}}{d^3k} = -\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^a(y)\rho^c(x)\,U^{de}(w')U^{eb}(w)\,\alpha$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^a(y)\rho^c(x)\,U^{db}(w)U^{ed}(x')\,\alpha$$
$$+\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)\,U^{db}(w')U^{ed}(x)\,\alpha'$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^c(x')\rho^a(y)\rho^e(x)U^{bd}(w')U^{ed}(w)\,\alpha'\Big] \tag{5.1}$$

$$\frac{d^3N_{b\,\rm NLO}}{d^3k} = \frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\,\log\left[\frac{\vee}{k^+}\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^h(y)\rho^e(x)U^{db}(w')U^{ec}(x)U^{ah}(y)\,\alpha$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{eb}(w)U^{ah}(y)\,\alpha'$$
$$-\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{be}(x)U^{ah}(y)\,\alpha'$$
$$+\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^e(x')\rho^h(y)\rho^d(x)U^{dc}(x)U^{be}(x')U^{ah}(y)\,\alpha\Big] \tag{5.2}$$

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[$$
$$-\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x)$$
$$+\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w)\,U^{ca}(x')\,\rho^c(x')\,\rho^b(x)$$
$$-\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$$
$$+\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$$
$$+\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ca}(x)\,\rho^b(x')\,\rho^c(x)$$
$$-\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)$$
$$-\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')$$
$$+\frac{1}{2}N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$$
$$-\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x)$$

$$
\begin{aligned}
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{dh}(x')\,\rho^{e}(x')\,\rho^{c}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{dh}(x)\,\rho^{c}(x')\,\rho^{e}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{dh}(w')\,U^{hb}(w)\,\rho^{a}(x)\,\rho^{e}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{dh}(x')\,\rho^{a}(x)\,\rho^{e}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{dh}(x')\,\rho^{a}(x)\,\rho^{e}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{bh}(w')\,U^{dh}(w)\,\rho^{a}(x)\,\rho^{e}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x')\,U^{al}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ec}(x')\,U^{db}(w)\,U^{al}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\Bigg]
\end{aligned}
\tag{5.3}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Bigg[\\
+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(w)\,U^{dh}(x)\,U^{ml}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^{m}(x')\,\rho^{e}(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ha}(w)\,U^{hl}(x')\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^{l}(x')\,\rho^{e}(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{ad}(z)\,U^{bh}(x)\,U^{el}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{ah}(x)\,U^{dl}(w')\,U^{eb}(z)\,\rho^{l}(x')\,\rho^{c}(x)\\
+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\frac{3}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w)\,U^{lh}(x')\,U^{da}(x)\,\rho^{l}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{hd}(x')\,\rho^{e}(x')\,\rho^{c}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^{e}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{hd}(x)\,\rho^{e}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(x')\,U^{lh}(x)\,U^{da}(z)\,\rho^{l}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{lc}(x')\,U^{be}(x)\,\rho^{h}(x)\,\rho^{l}(x')\Big]
\end{aligned}
\tag{5.4}
$$

6 The three-ρ terms cancel

Same test as in `LL_terms.pdf`, Sec. 7: every three-ρ term of the BFKL list (your lines, the blue lines, (B), and your d^3N_a, d^3N_b) is renamed to the frame w, w', z, charges x, y, x'; summed over the 6 placements of the charges and, for Groups I and III, its c.c.; random points, random SU(3) adjoint Wilson lines ($U^{bc} = U^{cb}$), random charges; $\log \frac{\vee}{k^+} = 1$. Your Group I Row II line 2 has the z integral done; it is written back as a z integral. Values at one configuration, in units of $i\mathcal{N}\,e^{-ik\cdot(w'-w)}$:

term	value
BFKL.tex G1 II(2a) line 1	-15.5494
BFKL.tex G1 II(2a) line 2	$+2.9492$
BFKL.tex G1 III N1 line 1	-1.6987
BFKL.tex G1 III N2 line 1	$+3.0986$
BFKL.tex G1 IV line 1	-1.8377
BFKL.tex G3 I line 1	$+1.8377$
BFKL.tex G3 II line 1	$+4.5784$
extracted G1 III N1 line 1	$+45.4745$
extracted G1 III N1 line 2	-45.4745
extracted G1 III N1 line 4	$+3.3973$
extracted G1 III N2 line 1	-19.6585
extracted G1 III N2 line 2	-32.9806
extracted G1 III N2 line 3	$+2.8850$
extracted G1 III N2 line 4	$+2.2837$
extracted G1 III N2 line 5	$+32.9806$
extracted G1 III N2 line 6	$+9.5968$
extracted G1 III N2 line 7	-21.5009
extracted G1 III N2 line 8	$+19.6585$
extracted G1 IV line 1	$+5.3403$
extracted G1 IV line 2	-9.9294
extracted G2 I(1) A line 1	$+1.9040$
extracted G2 I(1) A line 2	$+1.5852$
extracted G2 I(1) A line 3	-1.8952
extracted G2 I(1) B line 1	-2.0102
extracted G2 I(1) B line 2	$+0.0433$
extracted G2 I(1) B line 3	$+5.2217$
extracted G3 I line 1	-5.3403
extracted G3 I line 2	$+9.9294$
extracted G3 II line 1	-4.4229
extracted G3 II line 2	$+5.0271$
(B) G3 III(1)	-3.6368
(B) G3 V	$+3.6368$
(B) G3 VI	-0.0000
BFKL.tex d3N_a	$+6.3001$
BFKL.tex d3N_b	-1.7932
total	$+1.3e-15$

Largest relative residual over three configurations: 3.5×10^{-16}. Your `BFKL.tex` alone (without the blue terms): 0.39.

7 The two-ρ terms: BFKL

Every row of the BFKL list above equals the $\log \frac{\vee}{k^+}$ list of `LL_terms.pdf` (4.4×10^{-17}), whose two-ρ terms were added up and compared with BFKL$\otimes$LO kernel by kernel. They are, row by row (genuine two-ρ terms and the commutator terms of the three-ρ terms, $[\rho^a(u), \rho^b(v)] = if^{abc}\rho^c(u)\delta(u-v)$; orange / purple: 168 / 162):

Group I, Row II (three ρ;).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{I.II} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x) \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x') \\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x) \\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')\Big]
\end{aligned}
\tag{7.1}
$$

Group I, Row III (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{I.III} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x') \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^l(x)\,\rho^h(x') \\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x') \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x) \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\Big]
\end{aligned}
\tag{7.2}
$$

Group I, Row IV (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{I.IV} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{7.3}
$$

Group II, Row I (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{II.I} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x') \\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^l(x)\,\rho^h(x') \\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x) \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x') \\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x') \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^l(x')\,\rho^h(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\Bigg]
\end{aligned}
\tag{7.4}
$$

Group III, Row I (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.I} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x) \\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')
\end{aligned}
$$

$$
\begin{aligned}
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\Big[
\end{aligned}
\tag{7.5}
$$

Group III, Row II (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.II} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^{b}(x')\,\rho^{e}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')
\end{aligned}
$$

$$
\begin{aligned}
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&-2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^b(x)\,\rho^c(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^c(x')\,\rho^e(x)\Big]
\end{aligned}
\tag{7.6}
$$

Group III, Row III (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.III} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^l(x')\,\rho^h(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\Big]
\end{aligned}
\tag{7.7}
$$

Group III, Row V (three ρ).

$$
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.V} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[
$$

$$
\begin{aligned}
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\Big]
\end{aligned}
\tag{7.8}
$$

Group III, Row VI (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.VI} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ca}(z)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{eh}(z)\,\rho^{d}(x)\,\rho^{c}(x')\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ca}(z)\,\rho^{b}(x)\,\rho^{c}(x')\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(x)\,U^{eh}(z)\,\rho^{d}(x')\,\rho^{c}(x)\Big]
\end{aligned}
\tag{7.9}
$$

Group II, Row I (genuine two ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{II.I^{\rm gen}} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x')\,\rho^{c}(x)\,\rho^{l}(x')\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\Big]
\end{aligned}
\tag{7.10}
$$

Group II, Row IV (genuine two ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{II.IV^{\rm gen}} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;\rho^a(x)\,\rho^a(x')\\
&-2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{cb}(x')\,\rho^a(x)\,\rho^c(x')\\
&-2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)\Big]
\end{aligned}
\tag{7.11}
$$

Group III, Row III, second part (genuine two ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.III^{\rm gen}} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x')\,\rho^c(x)\,\rho^l(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x)\,\rho^l(x)\,\rho^h(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&-4\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{cb}(x')\,\rho^a(x)\,\rho^c(x')\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x)\,\rho^d(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x')\,\rho^d(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\Big]
\end{aligned}
\tag{7.12}
$$

From four ρ's: remainders d^3N_a, d^3N_b.

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{4\rho,\;a\text{-}b} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\Bigg]
\end{aligned}
\tag{7.13}
$$

From four ρ's: the two-ρ lists of `4rho.tex`.

$$
\begin{aligned}
\frac{d^3N}{d^3k}\bigg|^{\log\frac{\vee}{k^+}}_{4\rho\text{-lists}} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x)\\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^m(x')\,\rho^e(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^m(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^m(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^l(x')\,\rho^e(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\\
&+\tfrac{3}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)
\end{aligned}
$$

$$
\begin{aligned}
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ad}(w)\, U^{cl}(x')\, U^{be}(x)\, \rho^{l}(x')\, \rho^{h}(x) \\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{ade}\, U^{hb}(x)\, U^{hd}(x')\, \rho^{e}(x')\, \rho^{c}(x)\Big]
\end{aligned}
\tag{7.14}
$$

7.1 The sum

kernel structure	class	independent colour structures	differences
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	charge–charge	12	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	charge–charge	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	charge–charge	13	0
$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	charge–charge	14	0
$\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$	BFKL only	14	0
$\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$	BFKL only	10	0
$\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$	BFKL only	13	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$	BFKL only	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$	BFKL only	13	0

Collected, the sum is BFKL⊗LO, your App. E formula, Eq. (8.4), before the charge average (orange / purple: 23 / 12):

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\text{BFKL}\otimes\text{LO}} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[
&+N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x) \\
&-N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x) \\
&-N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x) \\
&+N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x) \\
&+N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x) \\
&-N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x) \\
&-N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x) \\
&+N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x) \\
&-2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{ha}(w)\, \rho^{b}(x)\, \rho^{e}(x') \\
&+2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dce}\, U^{hd}(x)\, U^{ha}(w')\, \rho^{b}(x')\, \rho^{e}(x) \\
&+2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{cde}\, U^{hd}(w')\, U^{ha}(w)\, \rho^{b}(x)\, \rho^{e}(x') \\
&+2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{ade}\, U^{hb}(x)\, U^{hd}(x')\, \rho^{e}(x')\, \rho^{c}(x) \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dbe}\, U^{ha}(w')\, U^{hd}(w)\, \rho^{c}(x)\, \rho^{e}(x') \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dbe}\, U^{ha}(w')\, U^{hd}(w)\, \rho^{c}(x')\, \rho^{e}(x) \\
&+2\,N_c\;\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;\rho^{a}(x)\,\rho^{a}(x') \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dbe}\, U^{ha}(w)\, U^{eh}(z)\, \rho^{d}(x)\, \rho^{c}(x') \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dbe}\, U^{ha}(w)\, U^{eh}(z)\, \rho^{d}(x')\, \rho^{c}(x) \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dbe}\, U^{ha}(w')\, U^{eh}(z)\, \rho^{c}(x)\, \rho^{d}(x') \\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dbe}\, U^{ha}(w')\, U^{eh}(z)\, \rho^{c}(x')\, \rho^{d}(x) \\
&-N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x) \\
&+N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x) \\
&+N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ca}(z)\,\rho^{b}(x')\,\rho^{c}(x) \\
&-N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ca}(z)\,\rho^{b}(x')\,\rho^{c}(x) \\
&-2\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{bde}\, U^{hd}(w')\, U^{ha}(w)\, \rho^{e}(x')\, \rho^{c}(x) \\
&-2\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{ha}(w)\, \rho^{b}(x)\, \rho^{e}(x') \\
&+2\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{ade}\, U^{hb}(w')\, U^{eh}(z)\, \rho^{d}(x)\, \rho^{c}(x') \\
&-2\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{ade}\, U^{hb}(x')\, U^{eh}(z)\, \rho^{d}(x)\, \rho^{c}(x') \\
&-N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;\tfrac{1}{(x'-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^{c}(x)\,\rho^{b}(x') \\
&+N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^{b}(x)\,\rho^{c}(x') \\
&+N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ca}(z)\,\rho^{b}(x)\,\rho^{c}(x') \\
&-N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ca}(z)\,\rho^{b}(x)\,\rho^{c}(x') \\
&-2\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{bde}\, U^{ha}(w')\, U^{hd}(w)\, \rho^{c}(x')\, \rho^{e}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^c(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(x)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\Big]
\end{aligned}
\tag{7.15}
$$

8 The result

Your commented "BFKL contribution" (both forms) is the sum, Eq. (7.15), to 2.1×10^{-16} (charges symmetrized, with c.c.):

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[
&N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x)\,\rho^e(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,\rho^a(x)\,\rho^a(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x)\,\rho^d(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x')\,\rho^d(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x)\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^b(x)\,\rho^c(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^c(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)
\end{aligned}
$$

$$+ 2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(x)\,U^{eh}(z)\,\rho^{d}(x')\,\rho^{c}(x)\Big] \tag{8.1}$$

$$\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\Big[\\
&- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) - N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\Big[\\
&- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+ N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) - N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
&+ 2\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) - 2\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)\\
&- 2\,f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x) + 2\,f^{abc}\,f^{dbe}\,U^{ha}(x')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)\Big]\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
&+ 2\,N_c\,\rho^a(x)\,\rho^a(x') - f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x')\\
&- f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x) + f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x)\,\rho^e(x') + f^{abc}\,f^{dce}\,U^{ha}(w')\,U^{hd}(z)\,\rho^b(x')\,\rho^e(x)\\
&+ f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x)\,\rho^e(x') + f^{abc}\,f^{dce}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x)\Big]\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\frac{1}{(x-z)^2}\Big[\\
&- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x) + N_c\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&- N_c\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\Big]\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
&+ 2\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') - 2\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x')\\
&- 2\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x') + 2\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(z)\,\rho^b(x)\,\rho^e(x')\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\frac{1}{(x'-z)^2}\Big[\\
&- N_c\,U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x') + N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^c(x)\,\rho^b(x')\\
&+ N_c\,U^{ab}(z)\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x') - N_c\,U^{ab}(z)\,U^{ac}(x)\,\rho^c(x)\,\rho^b(x')\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big[\\
&+ 2\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x) - 2\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(x)\,\rho^b(x')\,\rho^e(x)\\
&- 2\,f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(w)\,\rho^b(x')\,\rho^e(x) + 2\,f^{abc}\,f^{cde}\,U^{ha}(z)\,U^{hd}(x)\,\rho^b(x')\,\rho^e(x)\Big]\Big]
\end{aligned} \tag{8.2}$$

In color singlet form, we get

$$\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{1}{N_c^2-1}\,\log\Big[\frac{\vee}{k^+}\Big]\int_{x,,z,w,w'} e^{-ik(w'-w)}\,\langle\rho^f(x)\rho^f(y)\rangle\Big[\\
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(x,x,z)\Big[-N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K(y,y,z)\Big[-N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\Big]\\
&+K(w,w',z)\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(z)-U^\dagger(w))(U(z)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(z)-U^\dagger(w))(U(x)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w')\cdot\mathcal{K}(w-z)\,K(y,y,z)\Big[-N_c{\rm Tr}\{(U^\dagger(z)-U^\dagger(w))(U(x)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,K(x,y,z)\Big[2N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\}\Big]\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,K(y,y,z)\Big[-N_c{\rm Tr}\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\}\Big]\Big]
\end{aligned} \tag{8.3}$$

where $K(x,y,z)=\frac{(x-z)\cdot(y-z)}{(x-z)^2\,(y-z)^2}$

This is your App. E formula,

$$\frac{dN}{d^2w\,d^2w'\,dy}\bigg|_{\rm BFKL} = -\frac{\alpha_s^2 N_c\,\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,z}\langle\rho^f(x)\rho^f(y)\rangle\Big[\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')K_D(x,y,z)\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}$$
$$-2\,\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')K(x,y,z)\mathrm{Tr}\{(U^\dagger(z)-U^\dagger(w))(U(z)-U(w'))\}$$
$$+\mathcal{K}(z-w)\cdot\mathcal{K}(y-w')\big(2K(x,y,z)-K(x,x,z)\big)\mathrm{Tr}\{(U^\dagger(z)-U^\dagger(w))(U(y)-U(w'))\}$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(z-w')\big(2K(x,y,z)-K(y,y,z)\big)\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\}\Big], \tag{8.4}$$

$K_D = K(x,x,z) - 2K(x,y,z) + K(y,y,z)$, $\delta Y = \log\frac{\vee}{k^+}$, $\mathcal{N}\delta Y \leftrightarrow \frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}$: term by term, your singlet form is this with $\mathcal{K}(z-w) = -\mathcal{K}(w-z)$ and $x \leftrightarrow y$ in the last two lines. It agrees with BFKL⊗LO to 8×10^{-15} (`Final_evolution_check.pdf`). Hence

$$\frac{d^3N_{\rm BFKL}}{d^3k} = \frac{dN}{d^2w\,d^2w'\,dy}\bigg|_{\rm BFKL}\otimes\int_{w,w'}e^{-ik\cdot(w'-w)}, \qquad \delta Y = \log\frac{\vee}{k^+}. \tag{8.5}$$

9 The two red lines

Your Group I Row V 2b UV line and Group II Row IV line 1 both carry $\log\frac{\vee}{k^+}$ times a pole bracket. With $E(r) = \frac{2}{\epsilon} + 2\gamma_E + \log\frac{r^2\mu^2}{4}$ and $\mathcal{S}$ the LO structure (renaming as in `Final_evolution_check.pdf`, Sec. 6.1), they are $-\frac{g^4N_c}{16\pi^4}\big[E(x-w)+E(x'-w')\big]\mathcal{S}$ (with c.c.) and $+\frac{g^4N_c}{8\pi^4}E(w-w')\mathcal{S}$, times $\frac{1}{(2\pi)^3k^+}\log\frac{\vee}{k^+}$. Pointwise: the poles cancel (relative 1.5×10^{-15}), the sum does not (0.23): it is $\frac{g^4N_c}{16\pi^4}\log\frac{(w-w')^4}{(x-w)^2(x'-w')^2}\mathcal{S}$. Neither line is in your rows (`BFKL.tex` + `Other_terms.tex` exceeds the row by exactly these lines, check A of Sec. 10), and BFKL⊗LO is reproduced without them.

10 Checks

Scripts in `bfkl_rows/`:

check	result
A. `BFKL.tex` + `Other_terms.tex` = row − JIMWLK, row by row (`check_rows.py`)	$\le 2.5\times10^{-9}$ ($= O(\Lambda)$, $\Lambda = 10^{-9}$) except G1 V 2b UV, G2 IV (the red lines)
terms with an unevaluated integral: identical text	all
B. BFKL list (black + blue, without red) = $\log\frac{\vee}{k^+}$ list of `LL_terms.pdf`	$\le 4.4\times10^{-17}$
C. remaining terms = `Finite_terms_rows.pdf`	$\le 3.4\times10^{-16}$
three-ρ terms cancel: final / your `BFKL.tex` alone (`three_bfkl.py`)	3.5×10^{-16} / 0.39
four-ρ two-ρ lists vs the checked ones (`compare_4rho.py`)	3.2×10^{-15}
your "BFKL contribution" vs the checked sum (`compare_final.py`)	2.1×10^{-16}
red lines: poles / sum (`pole_bfkl.py`)	1.5×10^{-15} / 0.23
two-ρ sum vs BFKL⊗LO (`../final_check/render.py`)	0 of 109 differ